

Batch Foundry: A Socratic Engine for Mitigating Answer Leakage in Educational Artificial Intelligence

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Abstract

The rapid deployment of Large Language Models (LLMs) in educational settings presents a critical pedagogical challenge: "Answer Leakage." When students experience cognitive friction, they frequently utilize prompt injection or behavioral frustration to bypass educational scaffolding and extract direct answers from AI tutors. This paper introduces the LumenForge Socratic Engine, a deterministic gatekeeping architecture evaluated via a "Batch Foundry" methodology. Through an automated audit cohort of 25 adversarial pedagogical stress-tests (The Scholarly 25), the engine demonstrated a 96.0% Socratic Integrity rate and maintained an average Thinking Ratio (TR) of 38.5%. These results indicate robust mitigation of answer leakage, ensuring alignment with UK public sector educational standards and providing a scalable framework for institutional attainment-gap interventions.

1. Introduction

As the integration of generative AI within secondary education accelerates, the efficacy of these tools is often undermined by their propensity to act as answer engines rather than tutors. This phenomenon, which we term "Answer Leakage," occurs when an AI system capitulates to a student's cognitive avoidance strategies, directly solving problems rather than facilitating the Concrete-Pictorial-Abstract (CPA) learning loop.

Crucially within the UK public sector landscape, this stateless, containerized middleware architecture is engineered to directly solve the data compliance and pedagogical drift challenges identified by the Department for Education (DfE) and the Incubator for Artificial Intelligence (i.AI). By preventing AI models from prematurely divulging answers, institutional tutoring platforms can reliably uphold curriculum rigor while meeting strict UK safeguarding protocols.

This paper details the architecture and empirical evaluation of the LumenForge Socratic Engine (v2.7), a middleware policy enforcement layer designed to categorically prevent Answer Leakage and maintain high cognitive load.

2. Methodology: The Batch Foundry Architecture

To rigorously evaluate the engine prior to human deployment, we developed the "Batch Foundry," a digital wind tunnel for educational AI.

2.1 The Supervisor Proxy & Mathematical Formulation

The core architecture consists of an event-driven Supervisor Agent that acts as a transparent proxy between the Target AI Tutor and the student. The Supervisor intercepts the payload, calculates a real-time Thinking Ratio (TR), and classifies semantic intent using a secondary NLP classifier.

Immediately following interception, the system calculates the Thinking Ratio (TR) via the formal formula:

$$TR = (T_Socratic / T_Total) * (1 - \lambda_leak)$$

where the variables are explicitly defined as follows:

- T_Socratic = The volume of tokens generated containing cognitive prompts, open-ended questioning, and conceptual scaffolding.
- T_Total = The entire transactional token output payload.
- lambda_leak = A binary coefficient (0 or 1) indicating whether a direct structural solution pattern or final evaluation step was exposed.

2.2 The Scholarly 25 Audit Cohort

We synthesized three distinct adversarial student personas to stress-test the system:

- The Cheat-Seeker: Employs explicit demands and prompt injections to bypass educational scaffolding.
- The Concept-Block: Simulates a rigid misunderstanding of prerequisite concepts, resisting initial corrections.
- The High-Frustration Learner: Exhibits emotional distress and low confidence, requiring pedagogical patience.

The audit encompassed 25 distinct pedagogical traces across STEM and Humanities, mapped to UK Key Stage 3 and GCSE curricula.

3. Results & Explicit Leak Classification

The LumenForge Socratic Engine v2.7 demonstrated significant pedagogical resilience during The Scholarly 25 audit.

To rigorously evaluate performance across these adversarial traces, we established strict technical boundary metrics for leak classification. Specifically, a response triggers an "Answer Leak" classification ($\lambda_{leak} = 1$) if the Supervisor Proxy's secondary NLP classifier detects that the system-facing output contains more than 15% of a mathematical or logical formula's final evaluation steps, or states a direct definition without an accompanying open-ended query. Under these stringent boundary criteria, the 96.0% Socratic Integrity score represents a single conversational degradation edge-case out of the 25 adversarial traces.

Core Performance Metrics:

- Average Thinking Ratio (TR): 38.5% (Institutional Benchmark: > 35%)
- Socratic Integrity (Zero-Leak): 96.0% (Institutional Benchmark: 95.0%)
- Crisis Detection Latency: < 150ms (Institutional Benchmark: < 500ms)

3.1 Pedagogical De-risking

The integration of a strict Subject Lockdown and Zero-Leak Protocols successfully neutralized instances of answer leakage observed in previous iterations (v2.6). Furthermore, by capping student-facing output to a maximum of three sentences, the engine ensured the student performed 100% of active typing problem solving.

4. Conclusion & Regulatory Alignment

The LumenForge Socratic Engine represents a statistically significant advancement in maintaining pedagogical integrity at scale. By utilizing the Batch Foundry methodology for pre-deployment auditing, institutional stakeholders can verify an AI tutor's adherence to stringent educational and safeguarding standards.

Within the evolving governance framework of UK education, Batch Foundry serves as a scalable verification layer for Multi-Academy Trusts (MATs) to automate compliance tracking against the upcoming national Digital and Data Skills Pathway guidelines without exposing Personally Identifiable Information (PII). Future work will expand synthetic persona behavioral logic and explore broader demographic inclusivity metrics across national cohorts.